Calibration

Equivalence-scale specification — drop-in revision of the draft's calibration section

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1 Calibration

This section calibrates the equivalence-scale specification of the model. Two changes relative to the model section of the circulated draft matter for the calibration, and since this note travels on its own I state the new objects explicitly. First, children no longer enter through the committed consumption and housing bundles $(\bar{c}(n,s), \bar{h}(n,s))$ of the draft. Within-period utility for a household with n dependent children at home is

$$u(c,h,n) = e(n)\,\frac{\left(c^{\alpha(n)}\,h^{1-\alpha(n)}\right)^{1-\sigma}}{1-\sigma} + \psi\,n, \qquad \alpha(n) = \alpha_0 - \left(\delta_{\text{jump}}\mathbf{1}\{n \geq 1\} + \delta_a n\right), \tag{1}$$

where $e(n) = ((2+0.7n)/2)^{0.7}$ is the equivalence scale of Scholz et al. (2006), imposed rather than estimated. Children thus raise the marginal value of the whole consumption–housing composite through $e(n)$ and tilt spending toward housing through $\alpha(n)$, both only while they live at home; households without children at home have the homothetic preferences of the childless. There are no minimum-consumption or minimum-housing requirements, which is what permits empirical income risk below.

Second, fertility is sequential rather than one-shot. Each period during the fertile ages, a childless household chooses between waiting and trying for a first birth; a household at parity $n < 3$ with a child at home chooses between stopping and trying for birth $n+1$. An attempt at age a succeeds with the fecundity probability

$$\pi_a = \max\left\{0,\ 1 - 0.02\,e^{0.134\,(a-18)}\right\}, \tag{2}$$

which declines with age and reaches zero at 45; the coefficients are anchored to Leridon's clinical conception tables, with the formal fit pending. Completed family size is therefore an outcome of starting age, biology, and sequential choices—nobody chooses a number of children. Both decisions carry type-1 extreme-value taste shocks, with scale κ_E on the entry decision and κ_C on the continuation decision. The asymmetry is deliberate: the timing of the first attempt is allowed to be substantially random, standing in for the circumstances that time family formation, while the continuation decision of an established family is close to deterministic and therefore governed by prices, income, and preferences.

Second, income risk is set at its empirical value rather than truncated. The committed bundles of the previous specification forced income risk far below empirical estimates to keep the bundles affordable; without them the restriction disappears. I impose the persistent wage process of Flodén and Lindé (2001), with annual persistence 0.9136 and innovation standard deviation 0.206 net of measurement error, passed through the log-linear tax function of Heathcote et al. (2017) with progressivity $\tau = 0.181$, which preserves the autoregressive structure and scales the after-tax innovation to 0.169. Household wealth statistics are measured on the beginning-of-period balance sheet, in which liquid wealth and housing refer to the same point in time, and bequest flows are measured at the death node after the period's saving decision.

1.1 Parameters Set Outside the Estimation

Table 1 summarizes the parameters assigned outside the estimation. Demographics, housing finance, transaction costs, housing menus, and supply elasticity are unchanged from the previous draft: a four-year period; entry, retirement, and maximum ages of 18, 66, and 85; survival from the 2023 Social Security period life table; an 18-year expected duration of dependent children; $\sigma = 2$; an annual real return of 2.0 percent, housing depreciation of 1.1 percent, property tax of 1.0 percent, selling cost of 6 percent, labor-income tax of 17.9 percent, and a financed share $\phi = 0.80$ (Greaney et al., 2025; Davis and Van Nieuwerburgh, 2015); owner sizes $\{2, 4, 6, 8, 10\}$ rooms and a 6-room rental cap from the matched ACS distributions; supply elasticity $\eta = 1.75$ (Saiz, 2010); entrant wealth from the PSID distribution of childless renters aged 18–24; and $\theta_n = 0$, so bequest utility does not vary directly with the number of children.

Three assigned objects are new. The equivalence scale is fixed at the Citro–Michael form used by Scholz et al. (2006) and Borella et al. (2023); the square-root household-size scale is the robustness alternative. The fecundity schedule declines with age and closes at 45, anchored to the four-year conception probabilities in Leridon's clinical tables; its formal least-squares fit is pending, as is the CPS measurement of the weight on the top family-size bin, provisionally 3.4 children. The income process is the Flodén–Lindé–HSV object described above; unlike the previous draft, it is no longer provisional, and the footnote conditioning all quantitative results on a truncated income variance is retired.

Table 1: Parameters set outside the estimation

Parameter	Value	Source or restriction
Model period; entry, retirement, maximum ages	4 years; 18, 66, 85	Model timing
Post-retirement survival	Age-specific	2023 Social Security period life table
Expected years with dependent children	18 years	Child-maturity restriction
Risk aversion σ	2	
Equivalence scale $e(n)$	$((2 + 0.7n)/2)^{0.7}$	Scholz et al. (2006), from Citro and Michael (1995)
Fecundity by age	Declining; zero from 45	Leridon anchors; formal fit pending
Family-size states	$0, 1, 2, 3^{+}$; top-bin weight 3.4	Literal parity; CPS weight provisional
Annual income persistence ρ_z	0.9136	Flodén and Lindé (2001)
Annual income innovation s.d. σ_z	0.169	Flodén and Lindé (2001) innovation 0.206, scaled by $1 - \tau$ with $\tau = 0.181$ from Heathcote et al. (2017)
Annual real return; depreciation; property tax	2.0%; 1.1%; 1.0%	Greaney et al. (2025)
Financed share ϕ; sale cost ψ^s	0.80; 6%	Davis and Van Nieuwerburgh (2015)
Labor-income tax τ_{pay}	17.9%	OECD U.S. average
Owner house sizes; largest rental unit	$\{2, 4, 6, 8, 10\}$; 6 rooms	Matched ACS distributions
Housing-supply elasticity η	1.75	Saiz (2010)
Entrant wealth distribution	Wealth-to-income draws	PSID childless renters aged 18–24
Bequest child shifter θ_n	0	Maintained restriction

1.2 Internally Estimated Parameters

I estimate the remaining 12 parameters internally using the same 15 moments as the previous draft. Table 2 reports the estimates from the current diagnostic calibration. Although the parameters are chosen jointly, each parameter now has either one primary moment or membership in an explicitly named block with as many moments as parameters, and I state the assignment for every parameter in turn.

Table 2: Internally estimated parameters (diagnostic calibration)

Parameter	Interpretation	Estimate
β_{annual}	Annual discount factor	0.9705
α_0	Consumption expenditure share, no children at home	0.7542
δ_{jump}	Expenditure-share shift at entry into parenthood	0.0373
δ_a	Expenditure-share shift per additional child	0.0242
ψ	Utility from children	2.8137
κ_E	Taste-shock scale, first-birth timing	49.38
κ_C	Taste-shock scale, continuation decisions	0.1806
χ_O	Utility premium from owner-occupied housing	1.0588
$\bar{H}$	Housing-supply scale	8.5226
θ_0	Strength of the bequest motive	0.0264
θ_1	Bequest curvature shift	0.0442
κ_T	Dispersion in tenure choices	0.0002

Saving and bequests. The discount factor and the two bequest parameters form a three-parameter block matched to three wealth moments. β_{annual} is chosen to match the level of lifecycle wealth accumulation, currently summarized by the late-life estate median; θ_0 governs the resources intentionally carried to death and is matched to the same late-life wealth level jointly with the older nonhousing-wealth share; θ_1 governs how the bequest motive varies across the estate distribution. In the planned target revision these three parameters map one-to-one to the aggregate wealth-to-earnings ratio, the annual bequest-flow-to-wealth ratio, and the late-life wealth dispersion ratio $p_{90}/p_{50} = 3.448$ measured in the PSID; the aggregate ratio awaits the same-definition reconstruction of its borrowed value.

Housing demand, tenure, and supply. The expenditure share α_0 is pinned by the housing-expenditure slope of childless renters: with Cobb–Douglas preferences the share of allocatable expenditure spent on housing is $1 - \alpha_0$, and the CEX expenditure coefficient for childless cash renters gives $\alpha_0 = 0.733$, so this parameter has a direct empirical counterpart and its estimate can be read against it. The two tilt parameters are a triangular block matched to the two family-housing moments: the rooms increment at the first child identifies the total shift at entry into parenthood, $\delta_{\text{jump}} + \delta_a$, and the rooms gap between larger and smaller families identifies δ_a alone. χ_O is matched to the prime-age ownership rate, κ_T to the residual variance of tenure transitions, and $\bar{H}$ to aggregate occupied rooms; these three are one-to-one.

Fertility. The fertility side is a three-parameter block: ψ, κ_E, and κ_C against completed fertility, childlessness, and—in the planned revision—the timing of first births. The level of completed fertility primarily disciplines ψ. Childlessness disciplines the dispersion of family outcomes, and with continuation noise small, large families reappear because parents who start early progress to third births whenever the economics favors it. The current 15-moment system contains no timing moment, so κ_E is disciplined only indirectly; its estimate at the boundary of the search region, reported below, is the visible consequence, and the mean age at first birth with the share of first births after 30 will be added to close exactly this gap. The realized progression rates between birth orders remain overidentifying outcomes.

1.3 Target Moments and Fit

Table 3 reports all 15 target moments and their model counterparts at the current diagnostic estimate.

Table 3: Target moments and model fit (diagnostic calibration)

Moment	Data	Model
Family size and family housing		
Completed fertility	1.918	1.875
Childlessness rate	0.188	0.155
Family ownership gap	0.168	0.072
Housing increment with the first child, rooms	0.664	0.874
Rooms gap, 3+ versus 1–2 children, ages 30–55	0.368	0.554
Tenure and housing consumption		
Ownership rate, ages 30–55	0.575	0.678
Renter mean rooms, no children at home, ages 30–55	3.805	4.348
Owner share with rooms ≥ 6, no children at home, ages 30–55	0.596	0.852
Owner–renter mean rooms gap, no children at home, ages 30–55	2.419	2.038
Ownership rate, ages 25–34	0.341	0.359
Ownership rate, ages 65–75	0.764	0.945
Mean occupied rooms, ages 18–85	5.780	5.937
Wealth and bequests		
Young childless-renter wealth / annual income, ages 25–35	0.179	0.467
Median total estate wealth / annual income, ages 76–84	6.501	6.475
Share with nonhousing wealth $\geq$ annual income, ages 65–75	0.608	0.612

The model now matches the two completed-fertility moments closely—completed fertility of 1.875 against 1.918 and childlessness of 0.155 against 0.188—which the one-shot specification could reach only by overshooting fertility, and which no configuration of the sequential specification reaches without the margin-specific shock scales. Roughly half of model childlessness arises from households that try and run out the biological clock, consistent with the postponement mechanism at the center of the paper. The late-life estate median and the older nonhousing-wealth share are closely matched, and young ownership is now slightly overstated rather than understated.

Three misses deserve emphasis, in the same spirit as the previous draft. First, young childless renters accumulate wealth equal to 0.467 of annual income against 0.179 in the PSID. Under empirical income risk young households hold buffer wealth; this is a structural property of honest risk, smaller than under the previous draft's truncated-risk calibration once measurement is done consistently, and I report it as a disclosed miss rather than a target. Second, old-age ownership is 0.945 against 0.764, the same late-life exit failure as in the previous draft; here it is compounded by the bequest and tenure-dispersion estimates collapsing toward the boundary of the search region (θ_0, θ_1, κ_T in Table 2), which removes the motives that would let older owners sell. Third, the family ownership gap is understated at 0.072 against 0.168. The expenditure-share tilts at their current estimates are too small to generate the full gap; the diagnostic scans show the gap rising monotonically in the tilts, so this margin is available but not selected by the current target weights.

The estimate $\kappa_E = 49.38$ sits at the boundary of its search region, and this has a substantive consequence rather than a cosmetic one: with first-birth timing that random, fertility stops responding to prices and transfers, and the policy experiments reported below move births by essentially zero. I therefore treat the current calibration as diagnostic. It establishes that the sequential architecture can reach the fertility margins that motivated it, but its policy elasticities are not usable until the timing moments discipline κ_E away from the boundary, and the late-life exit margin and the family ownership gap require the revised target weights before the corresponding conclusions can be drawn.

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